2 A plane is flying with a velocity of 240 m s^{-1} at an angle of 30° with respect to the horizontal, as shown in Fig. 2.1. At an altitude of 2.0 km, a projectile is released from the plane. The projectile hits the target on the ground. Assume air resistance is negligible.

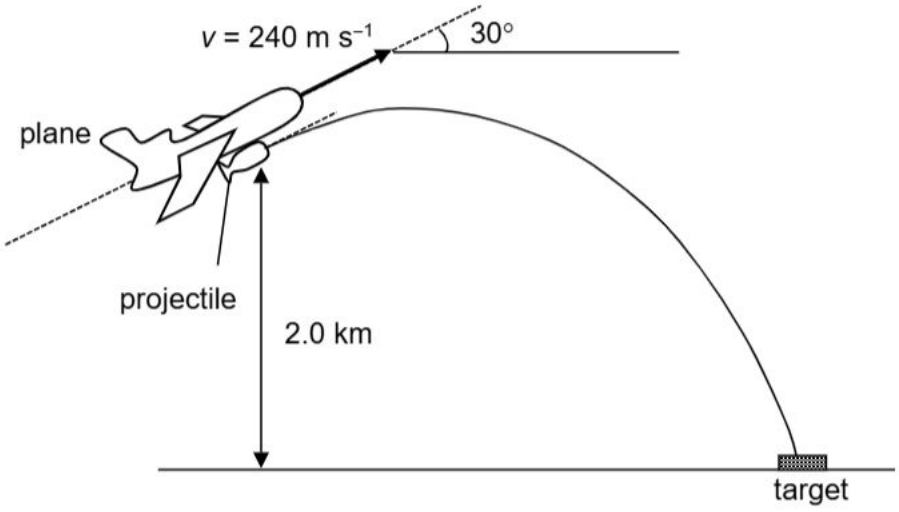


Fig. 2.1

(a) Determine the horizontal displacement of the projectile from the point of release to the target.

$$s_y = u_y t + \frac{1}{2} a_y t^2$$

$$-2000 = (240\sin 30)t + \frac{1}{2}(-9.81)t^2 \quad \text{[C1]}$$

$$4.905t^2 - 120t - 2000 = 0$$

$$t = -11.38 \text{ (reject)}, 35.8413 \text{ s}$$

$$s_x = u_x t$$

$$= (240\cos 30)(35.8413) \quad \text{[C1]}$$

$$= 7400 \text{ m} \quad \text{[A1]}$$

horizontal displacement = ... m [3]

(b) Determine the speed of the projectile just before it hits the target.

$$v_y = u_y + a_y t$$

$$= (240\sin 30) + (-9.81)(35.84)$$

$$= -231.6$$

$$v = \sqrt{v_x^2 + v_y^2}$$

$$= \sqrt{240\cos 30^2 + 231.6^2} \quad \text{[C1]}$$

$$= 311.2 \approx 310 \text{ m s}^{-1} \quad \text{[A1]}$$

OR

$$\text{KE}_{final} = \text{KE}_{initial} + \text{GPE}_{loss}$$

$$\frac{1}{2} m v_f^2 = \frac{1}{2} m(240)^2 + m(9.81)(2000) \quad \text{[C1]}$$

$$v_f = 311.2 \approx 310 \text{ m s}^{-1} \quad \text{[A1]}$$

speed = ... m s^{-1} [2]

(c) (i) Without further calculation, on Fig. 2.2, sketch the variation of vertical velocity v_y with time t of the projectile from the time it was released to the time it hits the target. Label this line **P**. Include values on your axes. [1]

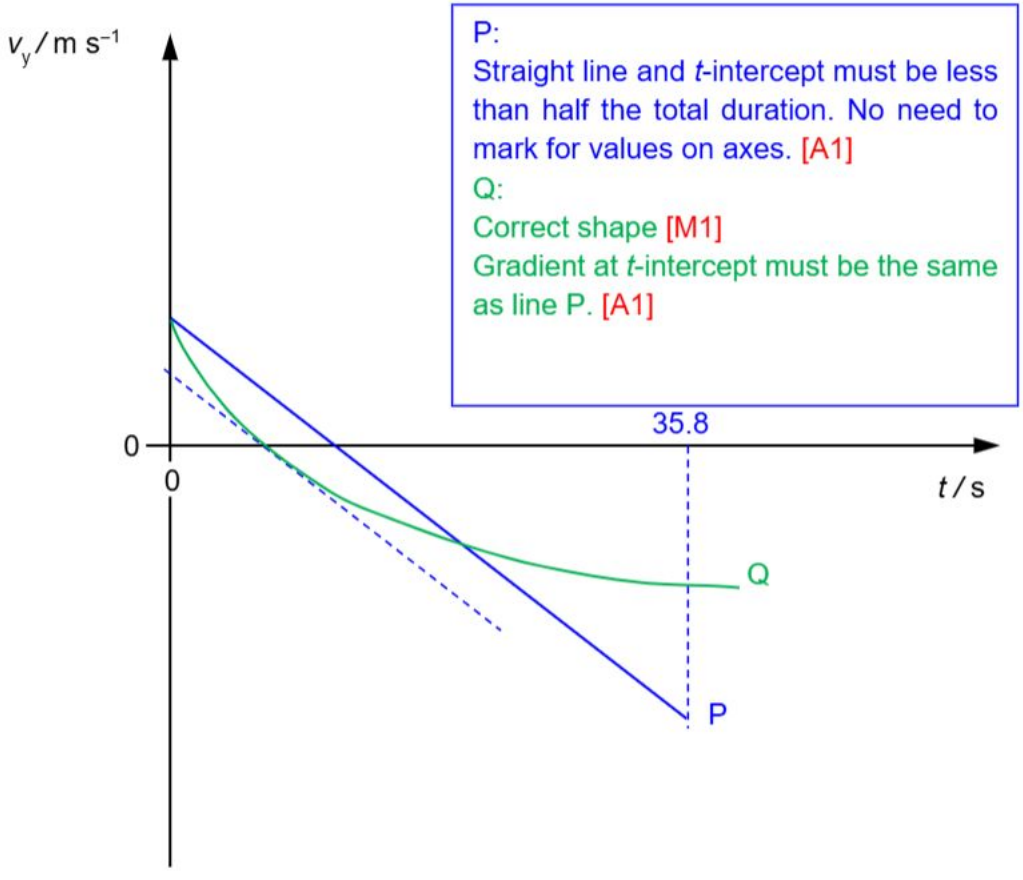


Fig. 2.2

(ii) On Fig. 2.2, sketch the variation of vertical velocity v_y with time t of the projectile if air resistance was **not** negligible. Label this line **Q**. No calculations are required. [2]

[Total: 8]